

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit 5 : Session 9 : SLO 1 | Instruction Formats (Zero/One/Two/Three-Address)

I. Multiple Choice

1. C) Stack zero-address instructions operate on the stack.
2. D) Accumulator the one-address format uses the accumulator as the implicit operand.
3. B) Destination operand in the two-address format the result overwrites the destination.

II. Fill in the Blanks

1. postfix zero-address format is ideal for postfix (reverse-Polish) expressions.
2. accumulator one-address format uses the accumulator for intermediate calculations.
3. three the three-address format explicitly specifies three operands (two sources + destination).

III. True or False

1. FALSE zero-address instructions take operands implicitly from the stack.
2. TRUE.
3. TRUE an explicit destination removes the need for temporary memory.

IV. Match the Following

- A) Zero Address 1) Stack-based.
B) One Address 2) Accumulator.
C) Two Address 3) Destination overwritten.
D) Three Address 4) Three explicit operands.

V. Numerical Answers

1. Zero-address code for $(A + B) * (C + D)$: PUSH A; PUSH B; ADD; PUSH C; PUSH D; ADD; MUL the stack machine leaves $(A+B)*(C+D)$ on top.
2. 4 bits $\log_2(16) = 4$ bits encode 16 registers.
3. 24 bits 8-bit opcode + 16-bit address = 24 bits per instruction.